

# SQL TRIGGERS

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# TRIGGERS

A trigger is a named database object that is associated with a table, and that activates when a particular event occurs for the table. The trigger becomes associated with the table named `tbl_name`, which must refer to a permanent table. You cannot associate a trigger with a `TEMPORARY` table or a view.

# TRIGGERS

Each trigger requires:

- A unique name. I prefer to use a name which describes the table and action, e.g. `blog_before_insert` or `blog_after_update`.
- The table which triggers the event. A single trigger can only monitor a single table.
- When the trigger occurs. This can either be `BEFORE` or `AFTER` an `INSERT`, `UPDATE` or `DELETE`. A `BEFORE` trigger must be used if you need to modify incoming data. An `AFTER` trigger must be used if you want to reference the new/changed record as a foreign key for a record in another table.

# TRIGGERS

Each trigger requires:

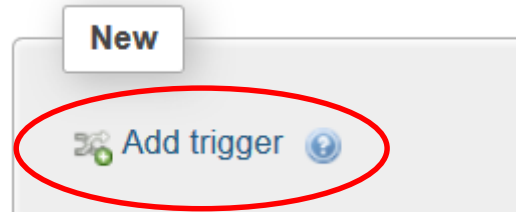
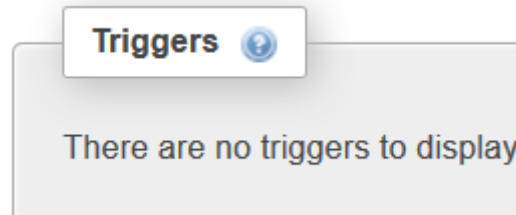
- The trigger body; a set of SQL commands to run. Note that you can refer to columns in the subject table using OLD.col\_name (the previous value) or NEW.col\_name (the new value). The value for NEW.col\_name can be changed in BEFORE INSERT and UPDATE triggers.

# TRIGGERS

Syntax:

```
CREATE TRIGGER `event_name`  
BEFORE/AFTER INSERT/UPDATE/DELETE  
ON `database`.`table`  
    FOR EACH ROW BEGIN  
        -- trigger body  
        -- this code is applied to every  
        -- inserted/updated/deleted row  
    END;
```

# TRIGGERS



# TRIGGERS

**Add trigger** ✕

Details

**Trigger name**

**Table**

filmy ▼

**Time**

BEFORE ▼

**Event**

INSERT ▼

**Definition**

1

**Definer**

Go

Close

# TRIGGERS

## Example 1

```
DELIMITER $$
CREATE TRIGGER `blog_after_insert` AFTER INSERT
ON `blog`
FOR EACH ROW BEGIN
    IF NEW.deleted THEN
        SET @changetype = 'DELETE';
    ELSE
        SET @changetype = 'NEW';
    END IF;
    INSERT INTO audit (blog_id, changetype) VALUES
    (NEW.id, @changetype);
END$$
```



# TRIGGERS

## Example 1

```
DELIMITER $$
CREATE TRIGGER `blog_after_update` AFTER UPDATE
ON `blog`
FOR EACH ROW BEGIN
    IF NEW.deleted THEN
        SET @changetype = 'DELETE';
    ELSE
        SET @changetype = 'EDIT';
    END IF;
    INSERT INTO audit (blog_id, changetype) VALUES
    (NEW.id, @changetype);
END$$
```

DELIMITER ;

# TRIGGERS

## Example 1

```
INSERT INTO blog (title, content) VALUES ('Article One',  
'Initial text.');
```

| id | title       | content      | deleted |
|----|-------------|--------------|---------|
| 1  | Article One | Initial text | 0       |

| id | blog_id | changetype | changetime          |
|----|---------|------------|---------------------|
| 1  | 1       | NEW        | 2011-05-20 09:00:00 |

# TRIGGERS

## Example 1

UPDATE blog SET content = 'Edited text' WHERE id = 1;

| id | blog_id | changetype | changetime          |
|----|---------|------------|---------------------|
| 1  | 1       | NEW        | 2011-05-20 09:00:00 |
| 2  | 1       | EDIT       | 2011-05-20 09:01:00 |

# TRIGGERS

## Example 1

UPDATE blog SET deleted = 1 WHERE id = 1;

| id | blog_id | changetype | changetime          |
|----|---------|------------|---------------------|
| 1  | 1       | NEW        | 2011-05-20 09:00:00 |
| 2  | 1       | EDIT       | 2011-05-20 09:01:00 |
| 3  | 1       | DELETE     | 2011-05-20 09:03:00 |

# TRIGGERS

## Example 2

A SIGNAL statement always specifies an SQLSTATE value, either directly, or indirectly by referring to a named condition defined with an SQLSTATE value. The first two characters of an SQLSTATE value are its class, and the class determines the default value for the condition information items:

- Class = '00' (success)
- Class = '01' (warning)
- Class = '02' (not found)
- Class > '02' (exception)

Example:

```
SIGNAL SQLSTATE '01234'
```

# TRIGGERS

## Example 2

Table of films.

```
DELIMITER $$
CREATE TRIGGER test_before_insert BEFORE INSERT
ON filmy
FOR EACH ROW
BEGIN
    IF CHAR_LENGTH( NEW.nazov ) < 4 THEN
        SIGNAL SQLSTATE '12345'
        SET MESSAGE_TEXT = 'GO HOME :>';
    END IF;
END$$
DELIMITER ;
```

# TRIGGERS

## Example 3

```
SELECT * FROM `t1`
```

|  |                                                                                   |      |                                                                                   |      | id                                                                                | text   | datetime                  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------|--------|---------------------------|
| <input type="checkbox"/>                                                          |  | Edit |  | Copy |  | Delete | 1 123 2017-04-11 11:52:34 |
| <input type="checkbox"/>                                                          |  | Edit |  | Copy |  | Delete | 2 22 2017-04-11 11:53:11  |
| <input type="checkbox"/>                                                          |  | Edit |  | Copy |  | Delete | 3 33 2017-04-11 11:53:32  |

# TRIGGERS

## Example 3

```
DELIMITER $$  
CREATE TRIGGER col_update  
AFTER UPDATE ON t1  
FOR EACH ROW BEGIN  
UPDATE t1 SET datetime=now() WHERE id=NEW.id;  
END$$
```

DELIMITER ;

|  |                                                                                     |      |                                                                                     |  |                                                                                      |        | id | text | datetime            |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|----|------|---------------------|
| <input type="checkbox"/>                                                          |   | Edit |   | Copy                                                                                |   | Delete | 1  | 123  | 2017-04-11 11:52:34 |
| <input type="checkbox"/>                                                          |  | Edit |  | Copy                                                                                |  | Delete | 2  | 22   | 2017-04-11 11:53:11 |
| <input type="checkbox"/>                                                          |  | Edit |  | Copy                                                                                |  | Delete | 3  | 33   | 2017-04-11 11:53:32 |



# TRIGGERS

## Example 3

```
DELIMITER $$  
CREATE TRIGGER col_update  
BEFORE UPDATE ON t1  
FOR EACH ROW  
BEGIN  
    IF OLD.id=NEW.id THEN  
        SET NEW.datetime=NOW();  
    end if;  
END$$  
  
DELIMITER ;
```